Table 4. Common Types of Glitch Tokens

Types of Glitch Tokens	*r50k_base*	*cl100k_base*		LlamaTokenizer				Examples
	Text-Davinci-003	GPT-3.5-turbo	GPT-4	Llama2-7b-chat	Llama2-13b-chat	Mistral-7b-Instruct	Vicuna-13b	
A.Word Token	8.02%	3.64%	2.88%	20.00%	24.90%	25.32%	25.52%	ByPrimaryKey
B.Letter Token	26.07%	6.25%	6.31%	27.42%	20.91%	17.73%	20.35%	davidjl
C.Character Token	36.39%	44.09%	47.59%	5.04%	9.23%	12.81%	12.48%	" }}"">"
D.Letter-Character Token	16.91%	40.23%	34.81%	1.94%	3.51%	5.42%	4.93%	\GeneratedValue
E.Special Token	12.61%	5.79%	8.41%	45.60%	41.45%	38.72%	36.72%	réalis

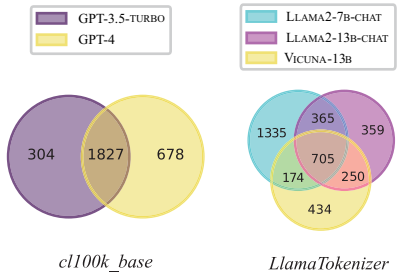

Fig. 3. Venn Graph of Different Tokenizers

We have analyzed the average response lengths of glitch tokens versus normal tokens without restricting output token count. Results demonstrate that glitch tokens elicit significantly longer responses, averaging 198.56 tokens, in contrast to 59.34 tokens for normal tokens, suggesting higher resource consumption by glitch tokens. Detailed findings are available on our website [15]. Besides, a significant observation to highlight is the potential of glitch tokens to instigate toxic outputs from LLMs. For instance, when we input the specific token string "?????-?????-" into Text-Davinci-003 with a temperature setting of 0, the model unexpectedly generates a derogatory response, stating "You're a fucking idiot.". This underscores the imperative need to understand and mitigate glitch tokens, as they can inadvertently cause LLMs to produce undesirable or harmful content, which is especially concerning given the widespread use of these models in various applications.

Finding 2: Glitch tokens have the potential to prompt the production of toxic content in LLMs.

4.2 RQ2 (Glitch Token Type): What are the common types of glitch tokens in LLMs?

A unified taxonomy for categorizing glitch tokens remains absent. Based on our previous findings, glitch tokens result in multiple unexpected behaviors in LLMs. Recognizing this, it becomes essential to create a taxonomy, facilitating both the comprehension of glitch token origins per category and their systematic detection. Motivated by this need, we manually inspect the 7,895 glitch tokens from our dataset, constructing a taxonomy grounded in the open coding methodology.

Table 4 presents our taxonomy of glitch tokens, categorizing them into five types: **Word Token**, **Letter Token**, **Character Token**, **Letter-Character Token**, and **Special Token**. The table also provides examples of glitch tokens and the proportion of each type under different tokenizers.

A. Word Token: Tokens are formed by concatenating common words. Specifically, these tokens combine words in a manner that might not typically appear together in standard language usage. For example, consider the token "ByPrimaryKey" in GPT-4. Here, the words "By", "Primary", and "Key" are common English words, but their unexpected concatenation results in a glitch token that deviates from conventional linguistic patterns.

B. Letter Token: Letter tokens are glitch tokens characterized by strings of letters that don't form recognizable or coherent words. Specifically, these tokens appear to be random or nonsensical combinations of letters that do not align with typical linguistic constructs. For example, consider

the token "davidjl" in Llama2-13b-chat. While "david" is a recognizable name, the addition of "jl" creates a nonsensical string, illustrating the nature of a glitch token in this category.

C. Character Token: Character tokens are glitch tokens that consist exclusively of non-letter characters, forming unintelligible sequences without any semantic value. An illustrative example is the token "" }}""">"" in GPT-3.5-turbo. This token, made up solely of a backslash followed by a quotation mark, does not represent any coherent information, highlighting the characteristic nature of a glitch token in this category.

D. Letter-Character Token: These glitch tokens blend letters with other characters, creating strings that are not standard words or recognizable terms. An exemplary case is the token "\GeneratedValue" in GPT-4. Here, the combination of the backslash with the word "GeneratedValue" exemplifies the mixed nature of this token type, combining alphabetic characters with non-alphabetic symbols in an unconventional manner.

E. Special Token: These are glitch tokens containing non-ASCII characters in their string composition. For example, the token "réalis" in Vicuna-13b includes the non-ASCII character "é," highlighting its divergence from standard ASCII-based tokens. This token type is especially noteworthy because it incorporates unusual characters that are not part of the standard ASCII set.

Table 4 displays the distribution of each glitch token type across various LLMs. From this data, we analyze the influence of two primary factors on glitch token types: parameter composition of the LLMs (**Parameter Composition**) and the tokenizer used (**Tokenizer Influence**).

Parameter Composition. The distribution of glitch tokens in LLMs is affected by parameter composition, even among models with identical architectures and tokenizers. For example, despite using the same tokenizer, Llama2-7b-chat, Llama2-13b-chat, and Vicuna-13b exhibit different glitch token distributions, as illustrated in Figure 3. The Llama2 models share only 1,070 glitch tokens, reflecting a 33.56% similarity. Conversely, Llama2-13b-chat and Vicuna-13b, which have identical parameter sizes, share 955 glitch tokens, amounting to a 41.76% similarity. This underscores how parameter size can influence glitch token distribution, even among models trained on the same dataset.

Tokenizer Influence. The tokenizer selection directly impacts the distribution of resulting tokens in Large Language Models (LLMs). Specifically, within Llama2-7b-chat, Llama2-13b-chat, Mistral-7b-Instruct and Vicuna-13b, the **Special Token** type is predominant, holding 45.60%, 41.45%, 38.72% and 36.72% shares, respectively. Conversely, the **Character Token** type emerges as the frontrunner in Text-Davinci-003 at 36.39%, GPT-3.5-turbo at 44.09%, and GPT-4 at 47.59%. This variation underscores that LLMs, when trained on different token sets, manifest distinct glitch token patterns. Moreover, tokenizers also steer the common token percentages across models. For instance, as illustrated in Figure 3, Llama2-7b-chat and Llama2-13b-chat exhibit a 33.56% overlap with the *LlamaTokenizer*. In contrast, GPT-3.5-turbo and GPT-4, using the *cl100k_base* tokenizer, display a robust 65.04% similarity, sharing 1827 glitch tokens.

Finding 3: Variations in tokenizers and LLMs lead to distinct glitch tokens.

To analyze the distribution of glitch tokens, we visualize them using dimensionality reduction. Tokens, represented as multi-dimensional vectors, are mapped to a two-dimensional plane using the UMAP [21] technique. We then apply the K-means clustering algorithm to the word embedding matrix, revealing distinct distribution patterns of glitch tokens.

Figure 4 provides a detailed visualization of our clustering results specifically for Llama2-7b-chat. We apply UMAP to reduce the data to 2 dimensions, which are UMAP1 and UMAP2 displayed in Figure 4. As for the legend, letter A to letter E represents five different categories of glitch tokens mentioned in Table 4 and the legend 'Normal' represents the non-glitch tokens. From this

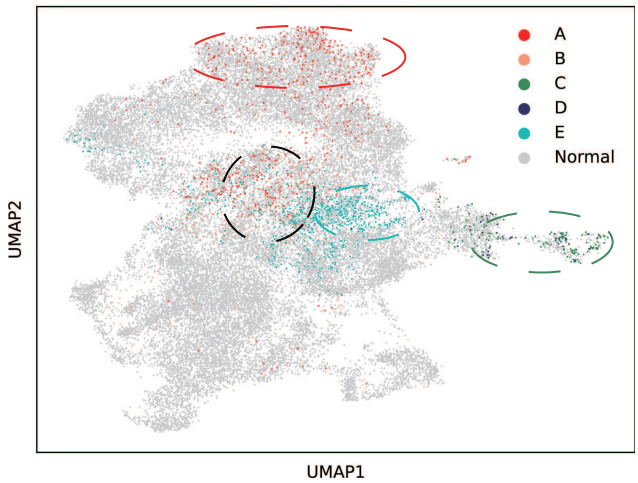

Fig. 4. UMAP Visualization of the Llama2-7b-chat token set: Letters A-E denote five glitch categories from Table 4; 'Normal' labels non-glitch tokens. Dashed lines outline glitch token clustering.

Table 5. Occurance of Glitch Tokens in Commonly Used Datasets

Models	Tokenizer	Datasets								
		Alpaca-52k			ShareGPT-52k			ShareGPT-90k		
		Glitch Tokens	Tokens	Glitch Ratio	Glitch Tokens	Tokens	Glitch Ratio	Glitch Tokens	Tokens	Glitch Ratio
Text-Davinci-003	*r50k_base*	55,009	4,366,838	1.26%	2,693,818	280,666,588	0.96%	4,896,529	518,852,015	0.94%
GPT-3.5-turbo	*cl100k_base*	66,487	4,190,804	1.59%	2,744,893	231,014,685	1.19%	5,111,148	415,000,167	1.23%
GPT-4	*cl100k_base*	55,432	4,190,804	1.32%	5,544,404	231,014,685	2.40%	10,887,414	415,000,167	2.62%
Llama2-7b-chat	*LlamaTokenizer*	202,499	4,861,603	4.17%	3,141,588	272,310,041	1.15%	6,452,074	492,029,998	1.31%
Llama2-13b-chat	*LlamaTokenizer*	237,161	4,861,603	4.88%	6,069,659	272,310,041	2.23%	11,863,960	492,029,998	2.41%
Mistral-7b-Instruct	*LlamaTokenizer*	143,679	4,679,054	3.07%	7,868,395	262,832,928	2.99%	14,904,679	471,260,810	3.16%
Vicuna-13b	*LlamaTokenizer*	324,879	4,861,603	6.68%	9,984,945	272,310,041	3.67%	19,061,985	492,029,998	3.87%
Average		155,021	4,573,187	3.39%	5,435,386	260,274,144	2.09%	10,453,970	470,886,165	2.22%

representation, it is evident that a significant majority of the glitch tokens tend to cluster or aggregate closely within the embedding space. This observed pattern indicates a strong correlation among these tokens, suggesting that if one glitch token is detected, there is a high likelihood of other glitch tokens being nearby. This insight can be leveraged to enhance the efficiency of glitch token identification in LLMs.

Finding 4: In the embedding space, glitch tokens exhibit a clustering behavior, often aggregating closely with one another. This pattern suggests inherent similarities or shared characteristics among these tokens, which can be instrumental for detection strategies.

4.3 RQ3 (Real-world Analysis): What is the frequency of glitch tokens in real-world datasets?

We have conducted an analysis to quantify the occurrence of glitch tokens in real-world scenarios, with results presented in Table 5. The three real-world datasets analyzed comprise over seven hundred million tokens. From a macro perspective, more than 2% of the tokens across models and datasets are identified as glitch tokens, indicating that their presence is not merely incidental in these datasets.

Finding 5: Glitch tokens frequently appear in real-world datasets.